



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

AGENTTRADE: A BULL-BEAR DEBATE DRIVEN MULTI-AGENT FRAMEWORK FOR STOCK ANALYSIS AND RISK-CONTROLLED PAPER TRADING

Team Members

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Problem Definition

- Stock research mixes price trends, news, portfolio limits and uncertainty.
- A chat model can answer confidently without showing evidence.
- Risk of an accidental jump from analysis to execution.
- Research, pending proposal and broker receipt are easily confused.
- Data outages can be disguised as a "missing stock".

Business / Real-World Context

- Stock Research-Helps users analyze market data, news, and supporting evidence.
- Decision Support-Provides structured insights and multiple perspectives before trading decisions.
- Paper Trading-Allows users to test trading decisions in a controlled simulation environment.
- Financial Research-Can support students, researchers, and retail users in studying market trends.
- Real-World Deployment-Requires stronger security, licensed financial data, operational controls, and expert/legal review.

Target Users

- Retail Investors
- Financial Researchers
- Stock Market Analysts
- Paper-Trading Users
- Investment Students

Stakeholder Identification

Stakeholder	Main Need
Retail Investor	Research stocks and evaluate recommendations
Financial Researcher	Analyze market data and supporting evidence
System Administrator	Manage models, data sources, and system configuration
Paper-Trading User	Review and approve paper-trading proposals

3 Proposed System: Analysis Pipeline

Transforms a stock-research question into an evidence-based analysis and risk-controlled paper-trading proposal through a multi-agent workflow..

1 Research Query

Accepts a stock-related question and identifies the required market information and evidence.

2 Retrieval Agent

Searches relevant financial data and retrieves supporting information from the knowledge base.

3 Analyst Agent

Analyzes market data, trends, indicators, and retrieved evidence to generate a recommendation.

4 Bull + Bear Agent.

Provides opposing perspectives by identifying potential opportunities, risks, and weaknesses.

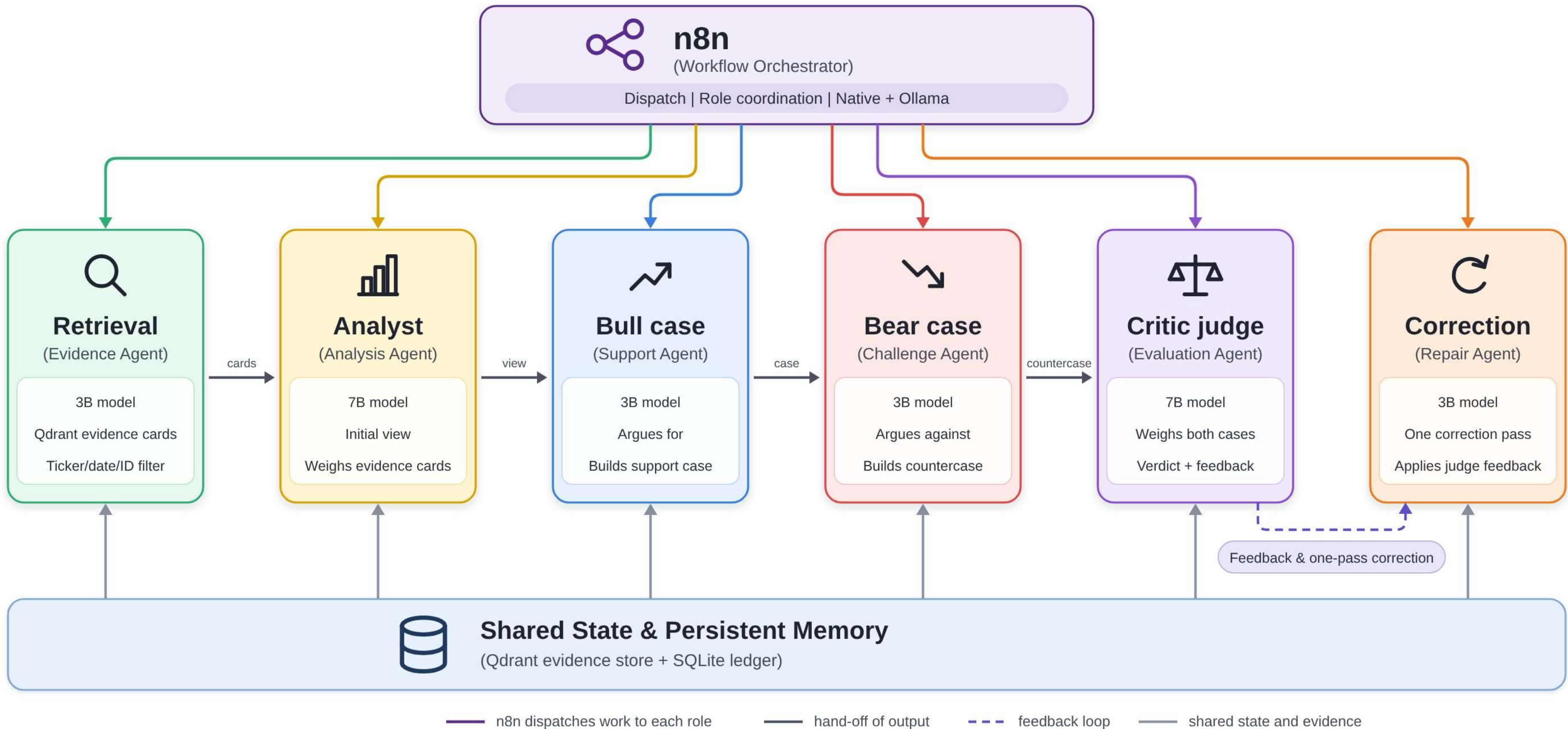
5 Critic+Correction

Critically evaluates the analysis and improves the response based on identified issues

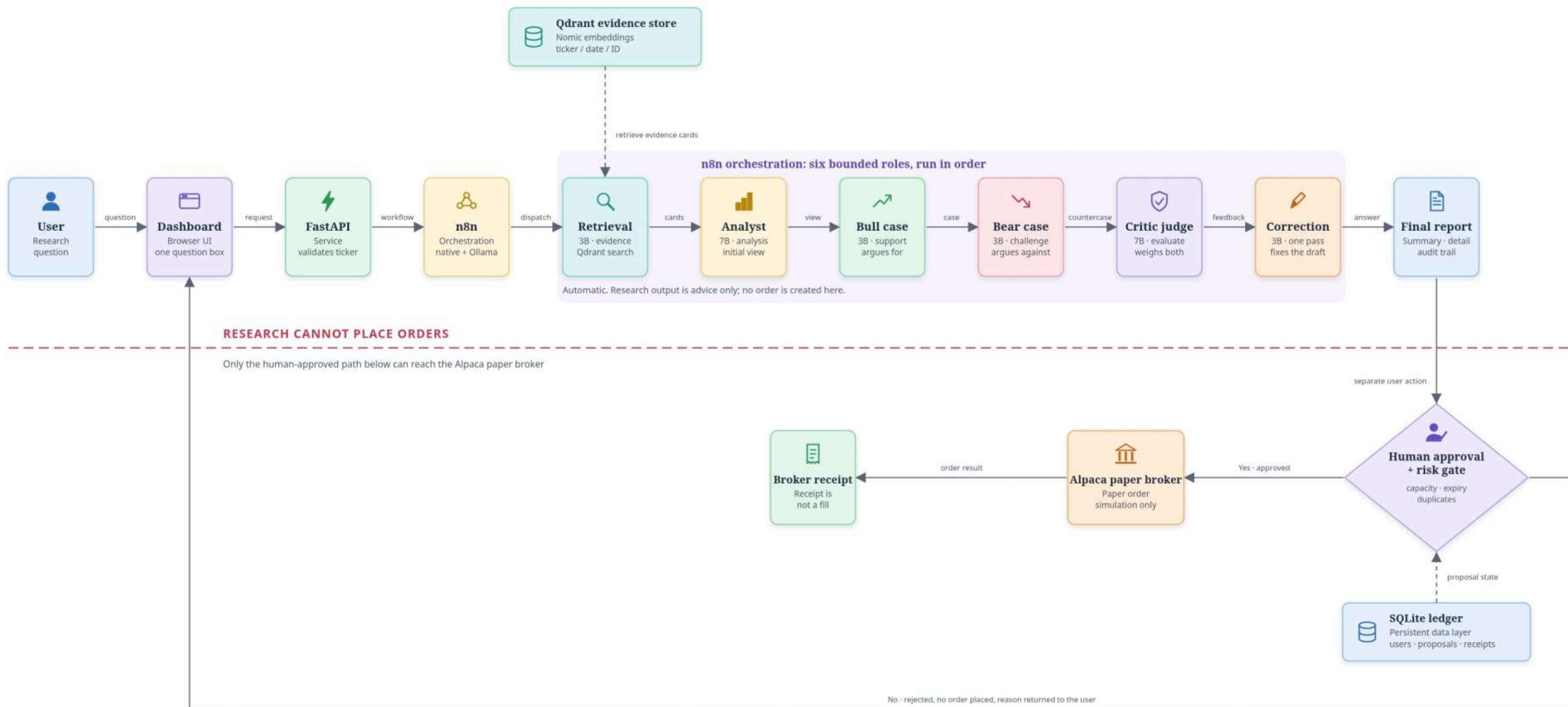
6 Risk Control + Approval

Applies deterministic risk limits and sends the proposal for human approval before paper-trading execution.

4 AGENTS ARCHITECTURE

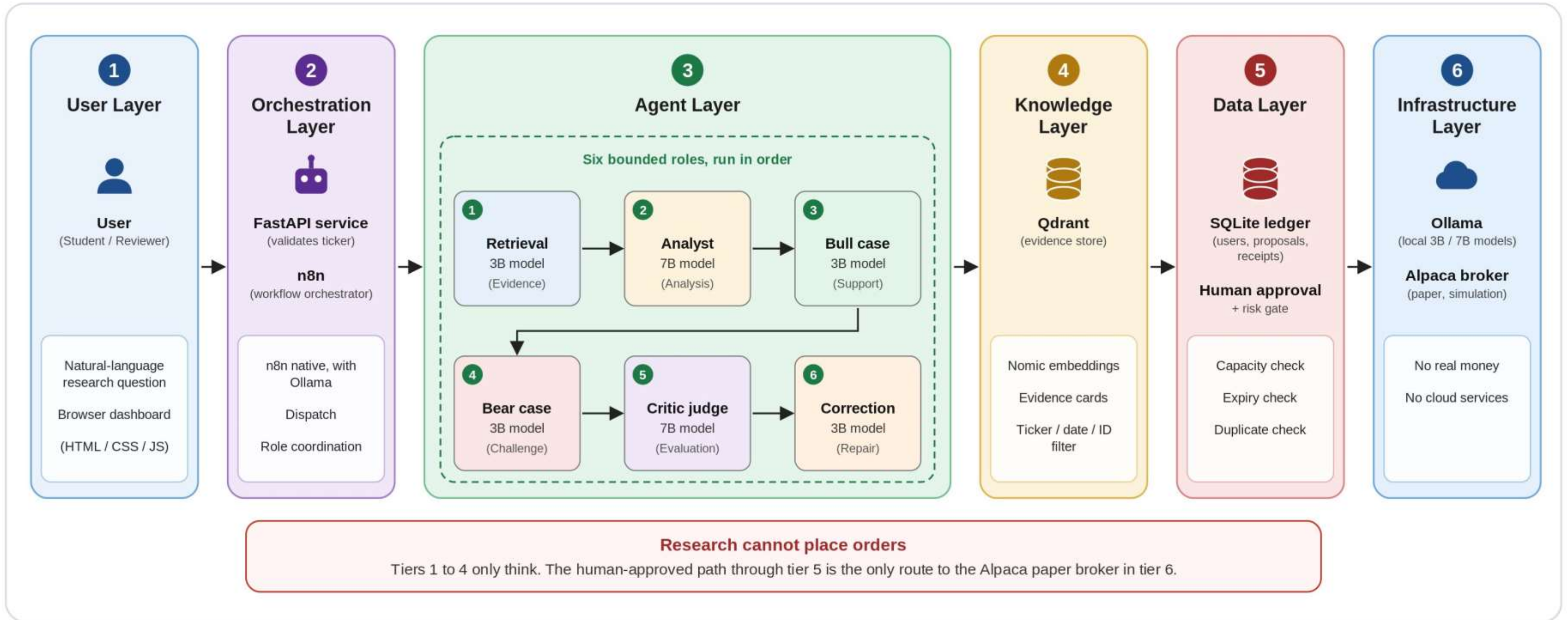


System architecture: automatic research path and human-approved paper-trading path



6

System Architecture: Tiered View



7 Proposed System: Data Sources

Provides financial market data, news, and knowledge resources required for stock research, retrieval, analysis, and paper-trading validation.

Yahoo Finance

Provides stock prices, market data, and technical indicators used for analysis.

NewsAPI

Provides financial news and supporting information for market research.

Frozen Synthetic Cards

Provides controlled benchmark evidence for testing retrieval and agent responses.

Qdrant Vector Database

Stores embedded evidence cards for semantic retrieval and Agentic RAG.

Alpaca Paper Trading

Provides a paper-trading environment for controlled order simulation and validation.

8 TOOLS AND TECHNOLOGIES USED

Layer	Technologies / Tools
1 User Interface Layer	 HTML  CSS  JavaScript  TradingView Widgets
2 API Layer	 FastAPI  Uvicorn  Pydantic  HTTPX
3 Agent & AI Layer	 Ollama  Qwen2.5 3B/7B  Llama 3.2 3B
4 Retrieval Layer	 Qdrant  Nomic Embeddings
5 Data & Orchestration Layer	 SQLite  n8n  yfinance  Pandas
6 Infrastructure & Trading Layer	 Docker Compose  Alpaca Paper API

Performance & Success Metrics

Important Metrics



1 Label-Term Accuracy

Measures expected-term matching in the generated answer.



2 Reasoning Proxy

Uses citation/rationale heuristics to assess reasoning.



3 Citation Rate

Measures detected citation syntax in the response.



4 Exact Consistency

Measures normalized equality between repeated answers.



5 Success / Errors

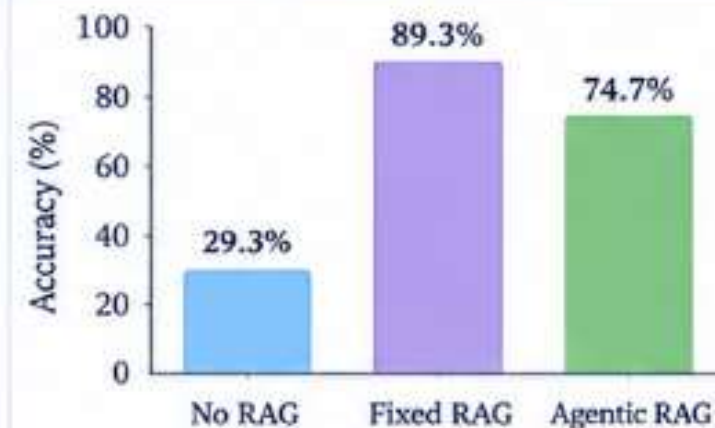
Measures workflow request completion and reported errors.

Performance Results

Qwen 3B



Llama 3B



Retrieval Mode	Qwen 3B	Llama 3B
No RAG	30.7%	29.3%
Fixed RAG	70.7%	89.3%
Agentic RAG	64.0%	74.7%



Key Finding: Fixed RAG achieved the highest reported label-term accuracy for both Qwen 3B and Llama 3B.

Success Condition



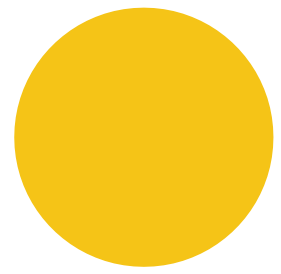
Main Result

AGENTTRADE successfully demonstrates a bounded multi-agent research workflow with evidence retrieval, critical debate, deterministic risk controls, and human-approved paper trading. The evaluation shows that retrieval improved reported accuracy over No-RAG, while Fixed RAG outperformed Agentic RAG on the tested benchmark.



Important

The performance values are reported results and were not independently rerun or verified.



Scalability & Real-World

Real-World Applicability

Financial Research – Supports structured stock research.

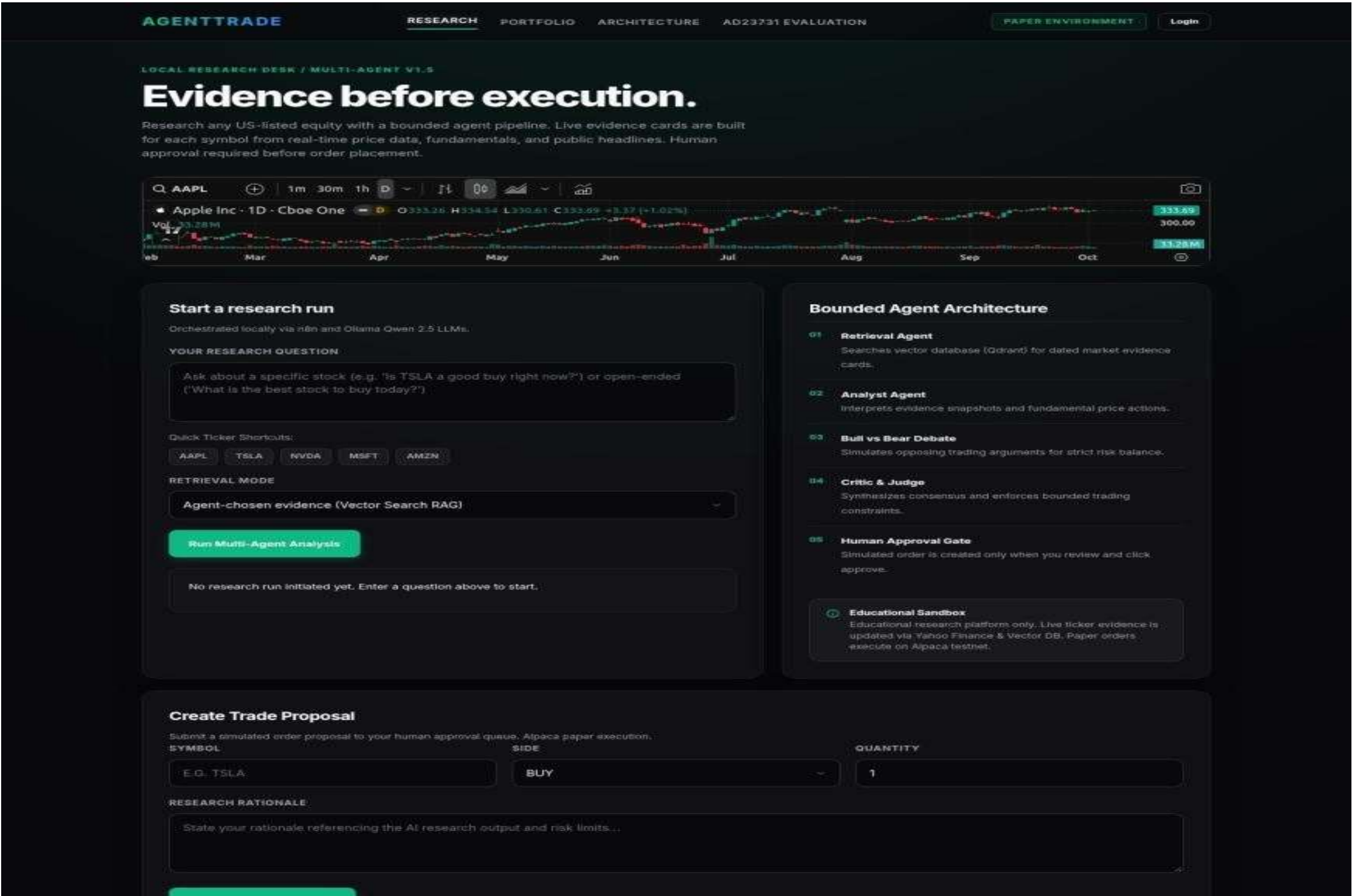
Paper Trading – Enables human-in-the-loop trading simulation.

Expert Review – Requires verified sources and expert validation.

Controlled Deployment – Real trading requires stronger security and operational controls.

Scalability

- **More Users** – Can be extended for wider user access.
- **More Data Sources** – Can integrate additional financial data sources.
- **Better Infrastructure** – Requires queues, authorization, and monitoring.
- **Higher Workload** – More candidates increase inference and data costs.





Result



AGENTTRADE

RESEARCH

PORTFOLIO

ARCHITECTURE

AD23731 EVALUATION

PAPER ENVIRONMENT

ALPACA PAPER / LIVE BROKER TELEMETRY

Paper portfolio

Broker-reported account values, open positions, and order ledger. Real-time paper execution.

Refresh portfolio

500

S&P 500 Index

7,721.9

+53.8 (+0.70%)

100

US 100 Cash

30,795.3

+286.9 (+0.94%)

Apple

333.69^D

+3.37 (+1.02%)

Tesla

173

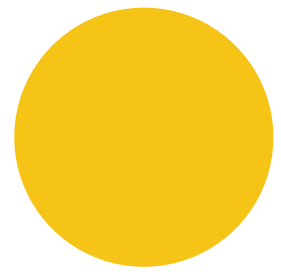
Paper account not configured.

Connect a paper account

Add Alpaca paper keys to show positions and equity history.

No sample holdings are displayed here.

Paper account only. Quotes and broker history may be delayed. Local sandbox endpoint; research proposals are managed on the [research desk](#).



CONCLUSION

- **AGENTTRADE** provides a bounded multi-agent framework for stock research and paper trading.
- **Evidence Retrieval** improves the quality of stock analysis compared with No-RAG.
- **Bull–Bear Debate and Critic Agents** provide multiple perspectives and improve transparency.
- **Risk Controls and Human Approval** keep paper-trading actions controlled and separate from research.
- The prototype demonstrates the potential of **multi-agent AI for structured and risk-aware financial research**



Thank You

Department of Artificial Intelligence and Machine Learning